26/03/26

Day 9.

Block 1 (Aptitude)

Time and Work

1) A's work / day = 1/10
B's work / day = 1/15

$\frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$

$\boxed{6 \text{ days}}$

2) A's 1 day work = 1/12
work done in 4 days = $\frac{4}{12} = \frac{1}{3}$
$1 - \frac{1}{3} = \frac{2}{3}$ (Remaining work)

(A+B) = $\frac{2}{3}$ in 4 days
1 day work = $\frac{2/3}{4} = 2/12 = 1/6$

$\frac{1}{12} \Rightarrow$ A's work

$\frac{1}{6} - \frac{1}{12} = \frac{1}{12}$ = B's work.

B's efficiency = $\frac{1}{12}$
B alone = 12 days

3) 5 people = 20 days

Total work = $5 \times 20 = 100$ units

10 days :

$100/10 \Rightarrow 10$ people

4)

A is twice efficient as B

B = 12 days

A = $12/2 = 6$ days

5) Pipe A = $1/6$

Pipe B = $1/8$

$$\frac{1}{6} + \frac{1}{8} = \frac{4+3}{24} = \frac{7}{24}$$

Time = $24/7$ hours

≈ 3.43 hours

$$\begin{array}{r} 3.429 \\ 7 \overline{)24} \\ 21 \\ \hline 30 \\ 28 \\ \hline 20 \\ 14 \\ \hline 60 \\ 56 \\ \hline 4 \end{array}$$

Block 2 (Verbal)

1) An algorithm is a step by step procedure used to solve a problem.

2) They improve performance by saving time and memory

3) They reduce time complexity and memory usage

Grammar correction

1) she <u>don't</u> like programming

she doesn't like programming

2) He <u>have</u> written the code.

He has written the code

3) They <u>is</u> working on algorithms

They are working on algorithm

Block - 3

When we run a program, several steps happen inside the computer.

First, the code is compiled or interpreted into machine code. Then memory is allocated in RAM for variables and data.

Next, the CPU executes the instructions step by step, performing calculations and logical operations

Finally, the program produces output, which is displayed on the screen